Hierarchical Model Specifications

Mathematical Formulations for WSLS, CFMR, and ECCM

Antigravity AI Pipeline

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Abstract

This document formally defines the mathematical architectures of the three models evaluated in the Hierarchical Bayesian parameter estimation pipeline. We detail the Win-Stay, Lose-Shift (WSLS) baseline, the Counterfactual Motor Representation (CFMR) baseline, and the ultimate Unified Cortico-Cerebellar Mixed Model (ECCM).

1 The Decision Rule: Drift Diffusion Model (DDM)

All three models utilize the Drift Diffusion Model (DDM) to generate reaction times (RT) and choices. The probability density function for a choice $C \in \{1, 2\}$ at time t is:

$$RT \sim \text{Wiener}(v^{(t)}, a, t_{nd}, w = 0.5) \tag{1}$$

Where a is the decision boundary, t_{nd} is the non-decision time, and $w = 0.5$ is the unbiased starting point. The distinguishing feature of each model is how it computes the drift rate $v^{(t)}$ on a trial-by-trial basis.

2 Model 1: Win-Stay, Lose-Shift (WSLS)

The WSLS model is a simple probabilistic tracking heuristic. It assumes the subject relies purely on the outcome of the immediately preceding trial.

2.1 Intuition

If you win, how likely are you to stay? If you lose, how likely are you to shift? This requires minimal memory and no gradual value learning.

2.2 Mathematics

Let p_{stay} and p_{shift} be the probability parameters. The drift rate towards choice 1 is:

$$v^{(t)} = \beta_v \cdot 2 \times \begin{cases} p_{stay} - 0.5 & \text{if previous choice was 1 and rewarded} \\ 0.5 - p_{stay} & \text{if previous choice was 2 and rewarded} \\ 0.5 - p_{shift} & \text{if previous choice was 1 and unrewarded} \\ p_{shift} - 0.5 & \text{if previous choice was 2 and unrewarded} \end{cases} \tag{2}$$

3 Model 2: CFMR (Counterfactual Q-Learning)

The CFMR represents the Cortical probability tracker. It learns the macroscopic reward rates of the environment using an asymmetric learning rate.

3.1 Intuition

The brain updates its belief about the chosen option based on whether it won or lost. Because it assumes a zero-sum environment, it counterfactually updates the unchosen option in the exact opposite direction.

3.2 Mathematics

Let Q_1 and Q_2 be the action values. The drift rate is driven by the value difference:

$$v^{(t)} = \beta_v \cdot (Q_1^{(t)} - Q_2^{(t)}) \tag{3}$$

Values are updated using asymmetric Long-Term Potentiation (LTP) and Depression (LTD):

$$Q_{chosen}^{(t+1)} = Q_{chosen}^{(t)} + \begin{cases} \eta_{LTP} \cdot (1 - Q_{chosen}^{(t)}) & \text{if rewarded} \\ \eta_{LTD} \cdot (-1 - Q_{chosen}^{(t)}) & \text{if unrewarded} \end{cases} \tag{4}$$

4 Model 3: ECCM (Parameterized Modulatory Cerebellum)

The ECCM is the ultimate integration of the CFMR Cortex and a high-dimensional Cerebellar Reservoir ($160 \to 1024 \to 256$).

4.1 Intuition

The massive Cerebellar network remembers the exact temporal sequence of the last 80 trials. It independently tries to predict the reward. If the Cerebellar prediction (Q_{CB}) agrees with the Cortical belief (Q_{CTX}), it multiplicatively amplifies the drift rate (faster reaction). If it disagrees (e.g., detecting a reversal before the Cortex realizes it), it suppresses the cortical signal to prevent a confident mistake.

4.2 Mathematics

The Deep Orthogonal Memory: Mossy Fibers (MF) hold the last 80 choices and 80 outcomes. Granule Cells (GC) project this into a 1024-dimensional space, and Molecular Layer Interneurons (MLI) provide 256-dimensional inhibition.

$$gc_j = \tanh\left(\sum W_{MF \to GC} \cdot MF\right) \tag{5}$$

$$mli_k = \tanh\left(\sum W_{GC \to MLI} \cdot gc\right) \tag{6}$$

$$Q_{CB} = \sum W_{PF} \cdot gc - \sum W_{MLI} \cdot mli \tag{7}$$

Thalamic Multiplicative Modulation: The Cerebellum projects to the Thalamus to modulate the Cortical drift rate:

$$v^{(t)} = \beta_v \cdot [Q_{CTX,1} \cdot (1 + w_{cb} Q_{CB,1}) - Q_{CTX,2} \cdot (1 + w_{cb} Q_{CB,2})] \tag{8}$$

This mathematically bounds the prediction and prevents Negative Log-Likelihood blowouts.